

3	(a) acceleration / force proportional to displacement from a fixed point acceleration / force (always) directed towards that fixed point / in opposite direction to displacement	M1	A1	[2]
(b)	(i) $A\rho g / m$ is a constant and so acceleration proportional to x negative sign shows acceleration towards a fixed point / in opposite direction to displacement	B1	B1	[2]
(ii)	$\omega^2 = (A\rho g / m)$ $\omega = 2\pi f$ $(2 \times \pi \times 1.5)^2 = (\{4.5 \times 10^{-4} \times 1.0 \times 10^3 \times 9.81\} / m)$ $m = 50 \text{ g}$	C1	C1	C1
		A1		[4]

4	(a) work done in bringing unit positive charge	M1
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